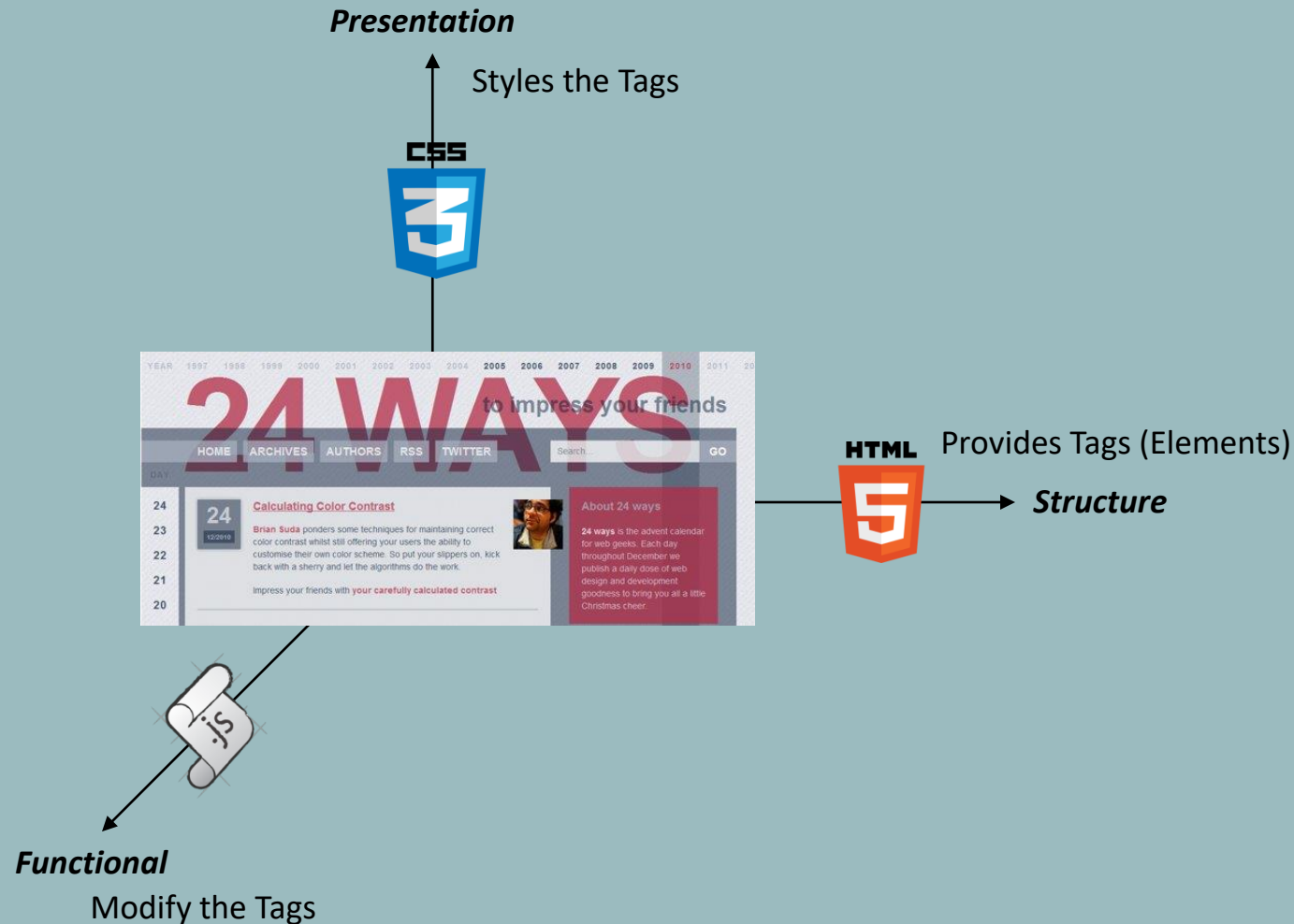


Web Development

>> CSS: Introduction

This lecture is based on lectures prepared by Dr. Mohammed Misbhaudhin | Updated and Modified by Dr. Mohamed Eltaweel

Separation of Concerns



Introduction - CSS

- Cascading Style Sheets
- Add Style & Layout to Webpages
 - Fonts, Colors, Spacing controlled by the CSS file
- Why CSS?
 - Separates **Content** from **Presentation**
 - Look-and-Feel can be updated independently

How to do it in CSS

`<h1> Welcome to the your first CSS Lecture </h1>`

HTML (inline)

`<h1 style="color:red"> Welcome to the your first CSS Lecture </h1>`

HTML with presentation

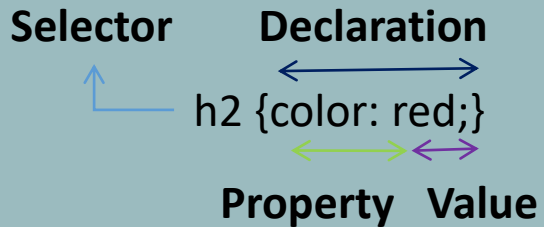
```
h1
{
    color: red;
}
```

CSS

The “CSS” Contents

- **Consists of CSS Rules**
- **Rules are composed of declarations**
 - Always put a semicolon after each declaration

Rule Structure



Types of CSS

- Inline
- Internal
- External



1. Inline :

- `<h1 style="color:red" > this is h1 heading </h1>`

Result: **this is h1 heading**

2. Internal CSS

```
<!DOCTYPE html>
<html>
<head>
<style>
body {background-color: powderblue;}
h1   {color: blue;}
p    {color: red;}
</style>
</head>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>

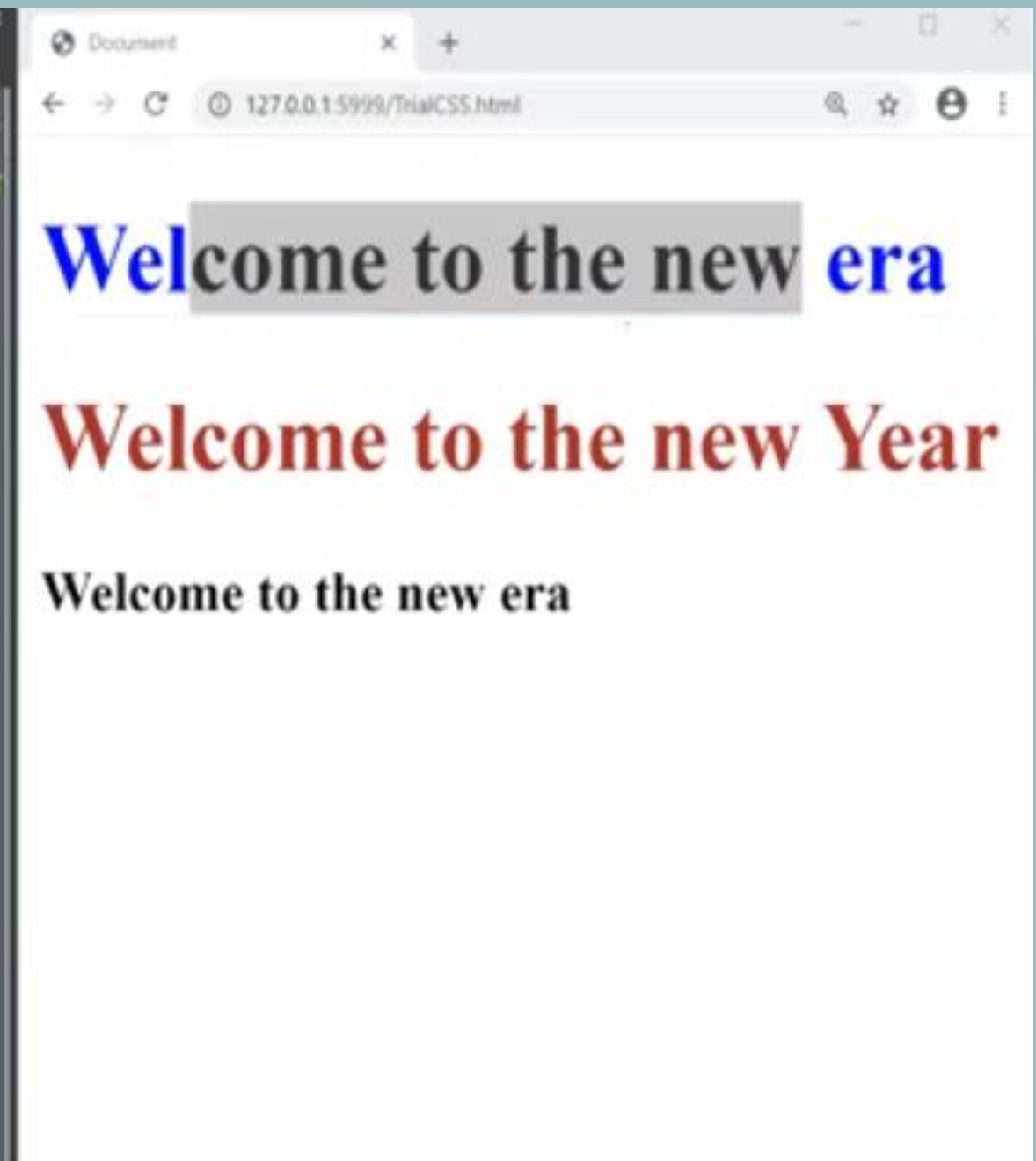
</body>
</html>
```



This is a heading

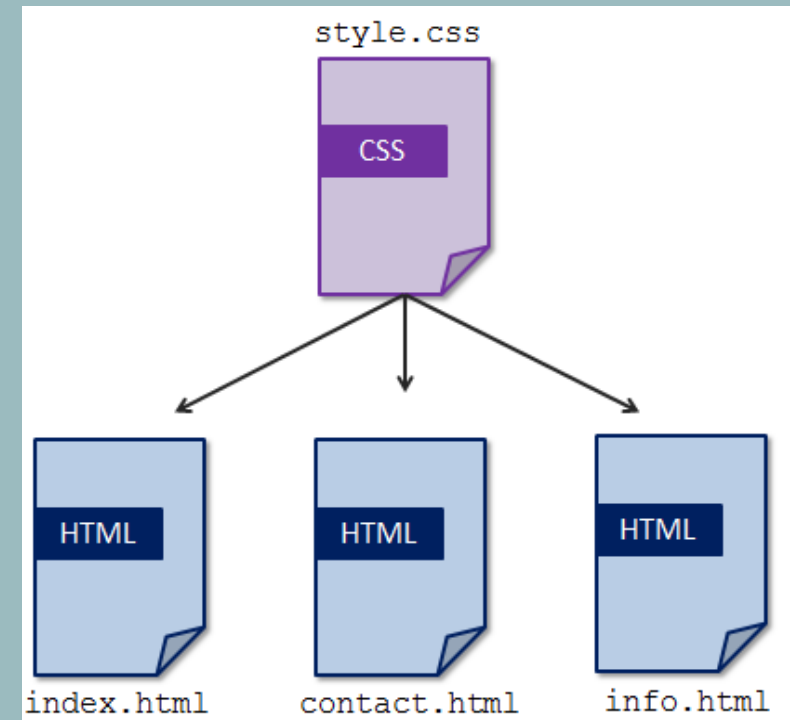
This is a paragraph.

```
4      <meta
      charset="UTF-8">
5
      <title>Document<
      /title>
6
      <style>
7      h1{color:
8      brown;}
9      </style>
10
11      I
11  </head>
12  <body>
13
14      <h1 style="color:
15      blue">Welcome to
16      the new era</h1>
17
18      <h1>Welcome to
19      the new Year</h1>
20  </body>
21  </html>
```



3. External CSS

- Used to define the style for many HTML pages.
- To use an external style sheet, add a **<link>** tag to it in the **<head>** section of each HTML page.
- *With an external style sheet, you can change the look of an entire web site, by changing ONLY one file!*

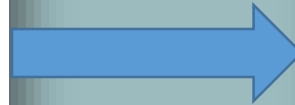


External CSS

```
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="styles.css">
</head>
<body>

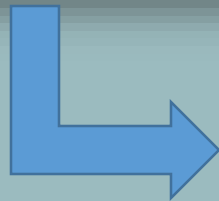
<h1>This is a heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```



This is a heading

This is a paragraph.



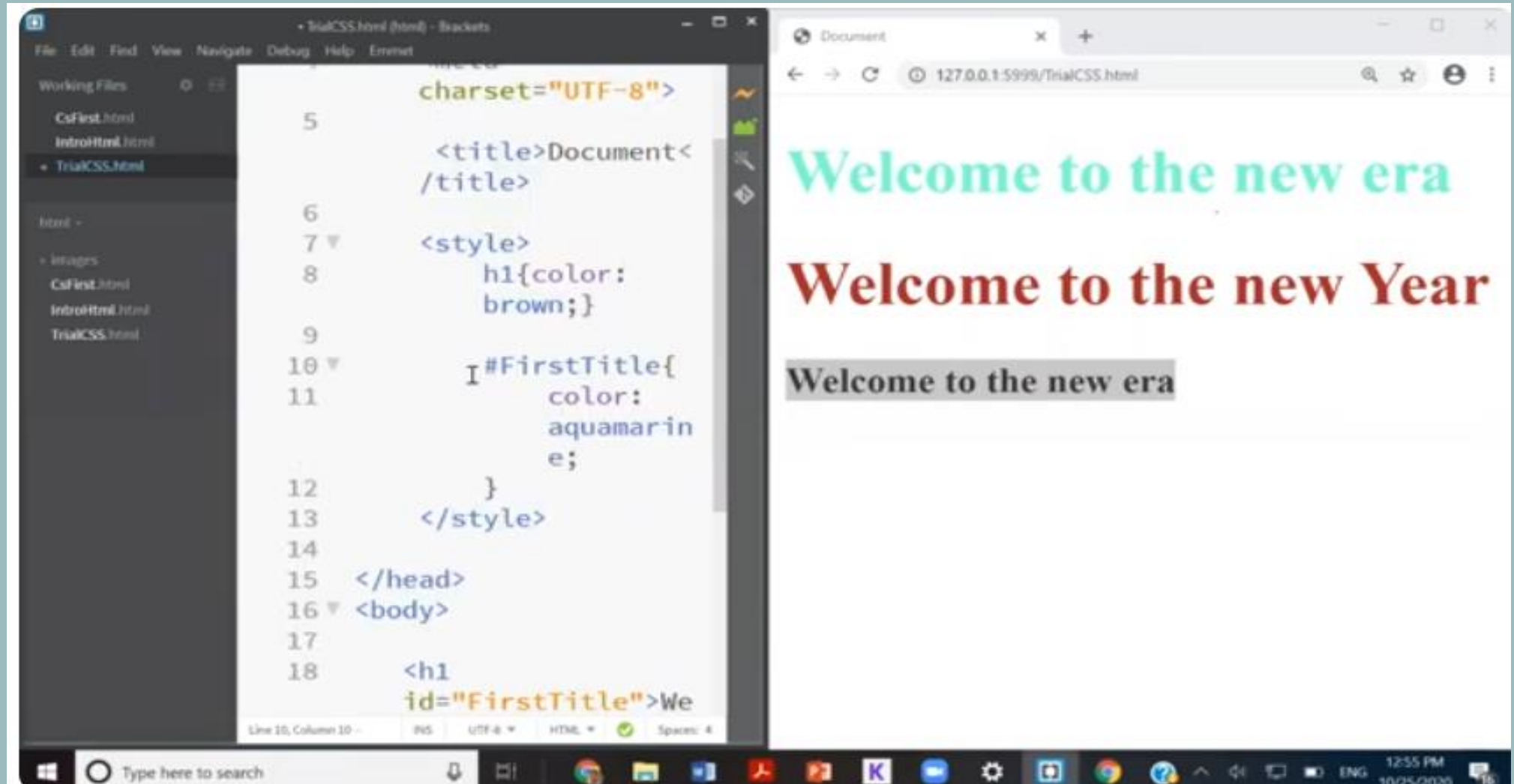
styles.css

```
body {
  background-color: powderblue;
}
h1 {
  color: blue;
}
p {
  color: red;
}
```

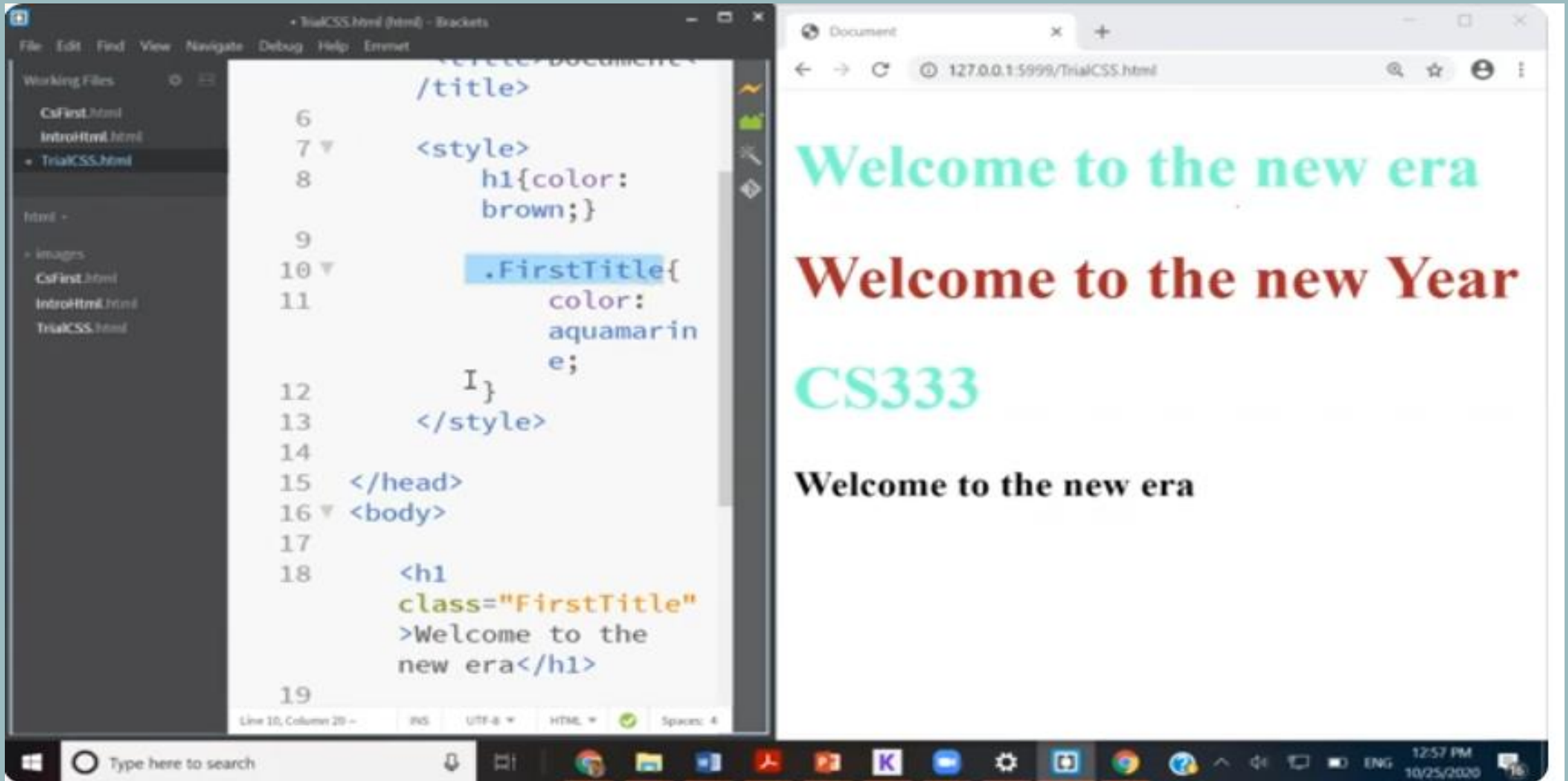
Attribute “id” & “class”

- Each element (tag) in HTML can have an “id” attribute and a “class” attribute
- Difference
 - “id” value is unique to a particular element
 - “class” value can be shared by multiple elements

grab the element by id (#)



grab the element by class name (.)



Adding CSS to HTML

- **There are three locations to add CSS to HTML**
 - We will talk about them later in the course
-

Steps to add CSS to HTML

- Create `<style> ..... </style>` element inside the `<head>` element
- Each CSS rule is written inside the `<style>` element
- This method is called Internal CSS

TRY NOW

1. Add a level 1 header with any text
2. Change the header's **color** is red, the **font family** is Georgia and the **font size** is 32px

HINTS:

Use properties

color
font-family
font-size

Selector Types in CSS

- **Element (Type) Selector**

- Just use the tag name in the CSS Selector field
 - `h1 {color: red;}`

- **ID Selector**

- Use “#” before the name of the id in CSS Selector field
- `#title {color: red;}` `<h1 id=“title”>Welcome</h1>`

- **Class Selector**

- Use “.” before the name of the class in CSS Selector field
- `.headings {color:red;}` `<h1 class=“headings”>Welcome</h1>`

TRY NOW

1. *Add two paragraphs to your page with any text*
2. *Now change the font color of the “second” paragraph to green*

HINTS:

*Assign id attributes to the
respective tag in HTML*

TRY NOW

1. *Add one more paragraph to your page*
2. *Now change the font size of the “first” and “third” paragraph to 10*

HINTS:

*Assign class attributes to
the respective tags in
HTML*

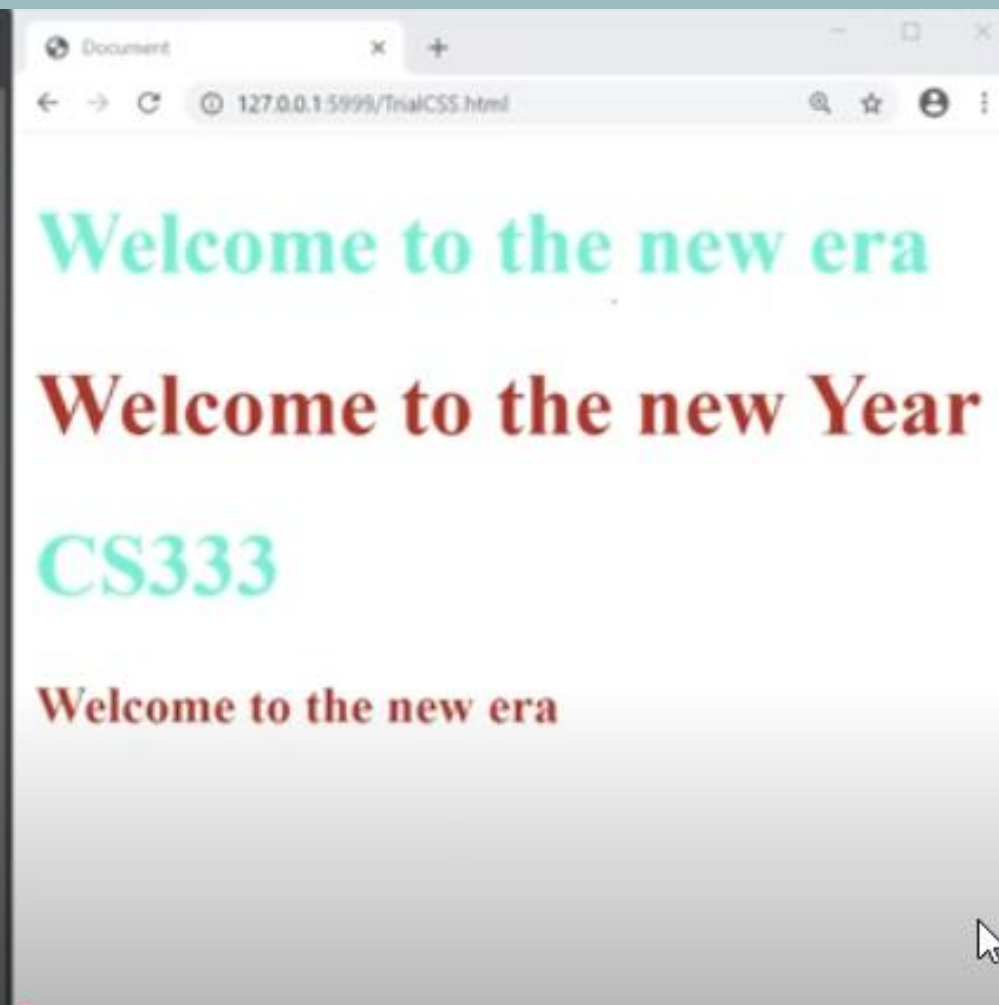
Combining Selectors

- **Apply same style to different selectors**
- **Use “comma (,)” to separate the selector names in the CSS Rule**

- Eg

```
h1, h2
{
    color: red;
}
```

```
• TrialCSS.html (html) - Brackets
File Edit Find View Navigate Debug Help Emmet
Working Files
  CsFirst.html
  IntroHtml.html
  • TrialCSS.html
html -
  Images
  CsFirst.html
  IntroHtml.html
  TrialCSS.html
4 <meta
  charset="UTF-8">
5
6 <title>Document<
  /title>
7 <style>
8   h1,h3{color:
  brown;}
9
10
11
12 .FirstTitle{
13   color:
  aquamarin
  e;
14 }
15 </style>
16
17 </head>
18 <body>
```

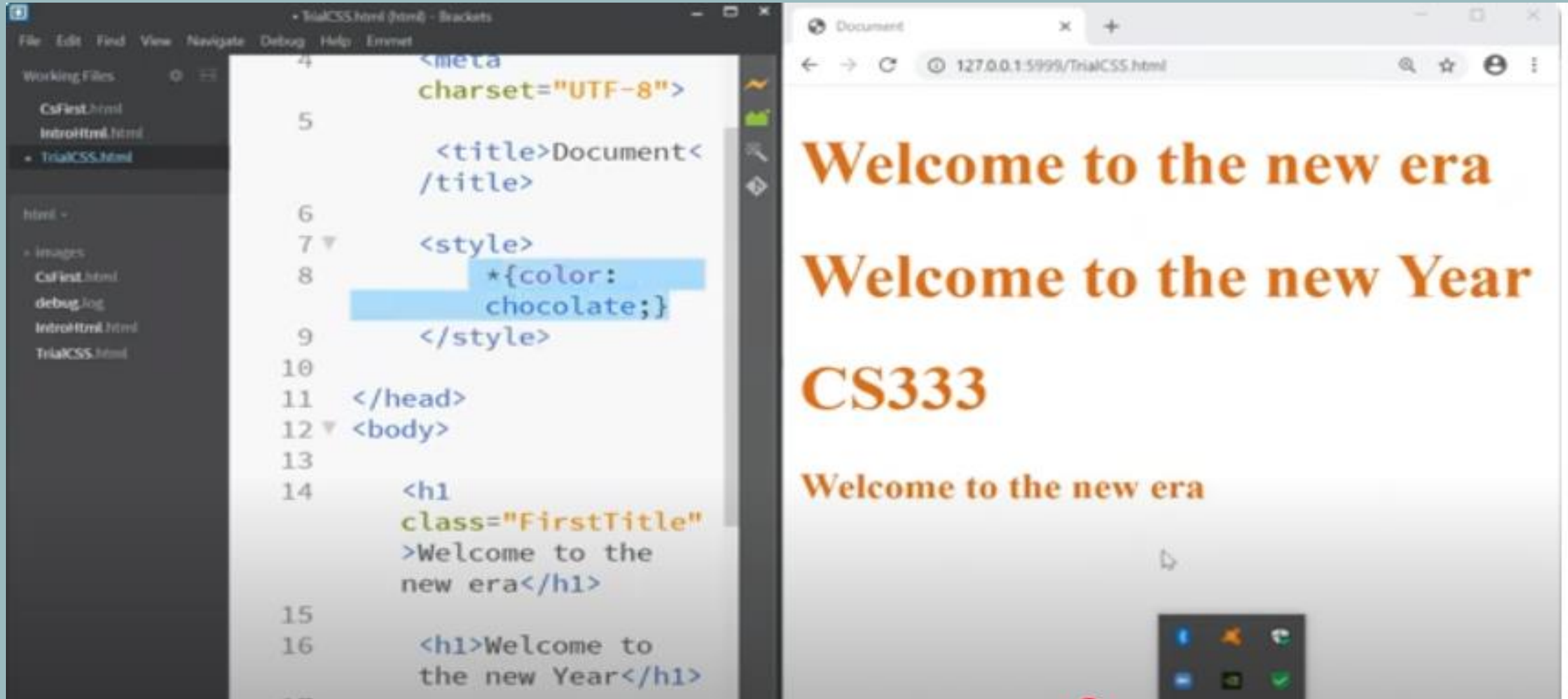


The Universal Selector

- The universal selector, denoted by an asterisk (*), applies its style to all elements in a document.

* { property-value list }

The Universal Selector



TRY NOW

1. *Center align the heading & the first paragraph*

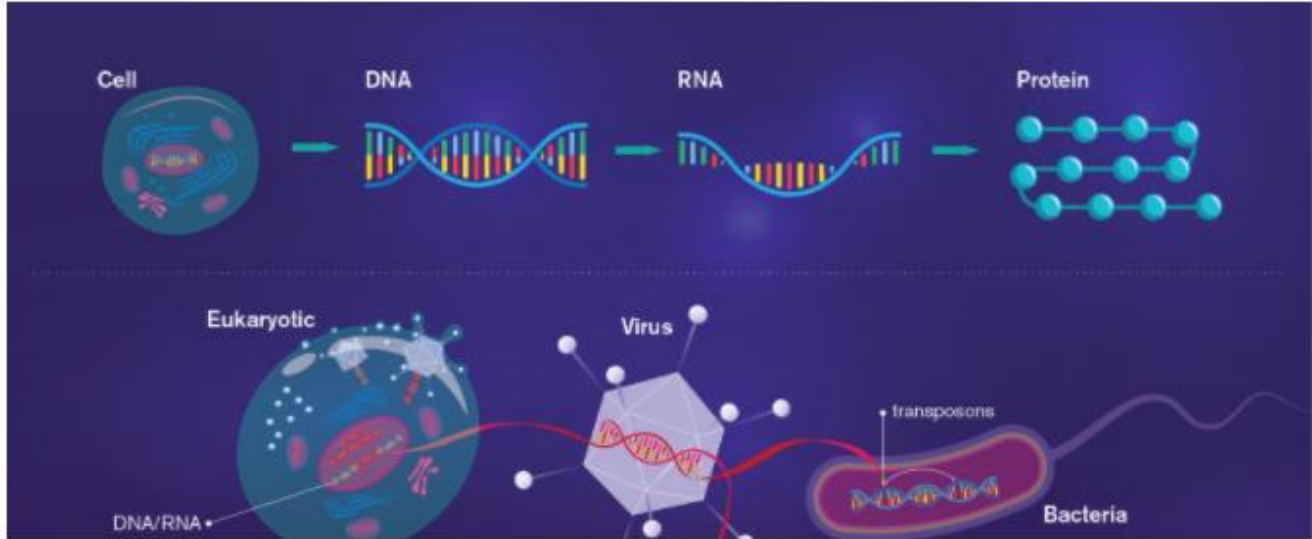
HINTS:

Use the property
text-align

Next on Web-based Systems

- **Cascading Style Sheets**
 - More on CSS Selectors
 - Using to format the layout of an HTML page
 - Using <div>, tags and CSS code
- **More on CSS Benefits**
 - <http://csszengarden.com>
- Let's show one of CSS ready templates

Preview your html (inspect)



The diagram illustrates the flow of genetic information. At the top, a sequence shows a **Cell** containing **DNA**, which is transcribed into **RNA**, and then translated into a **Protein** (represented as a chain of blue spheres). Below this, a more detailed view shows **Eukaryotic** cells with **DNA/RNA** inside, a **Virus** with its own DNA, and **Bacteria** containing **transposons** (mobile genetic elements). Arrows indicate the movement and interaction of these genetic components.

Below the diagram is a browser's developer tools interface. The **Elements** panel shows the HTML structure, with the following code visible:

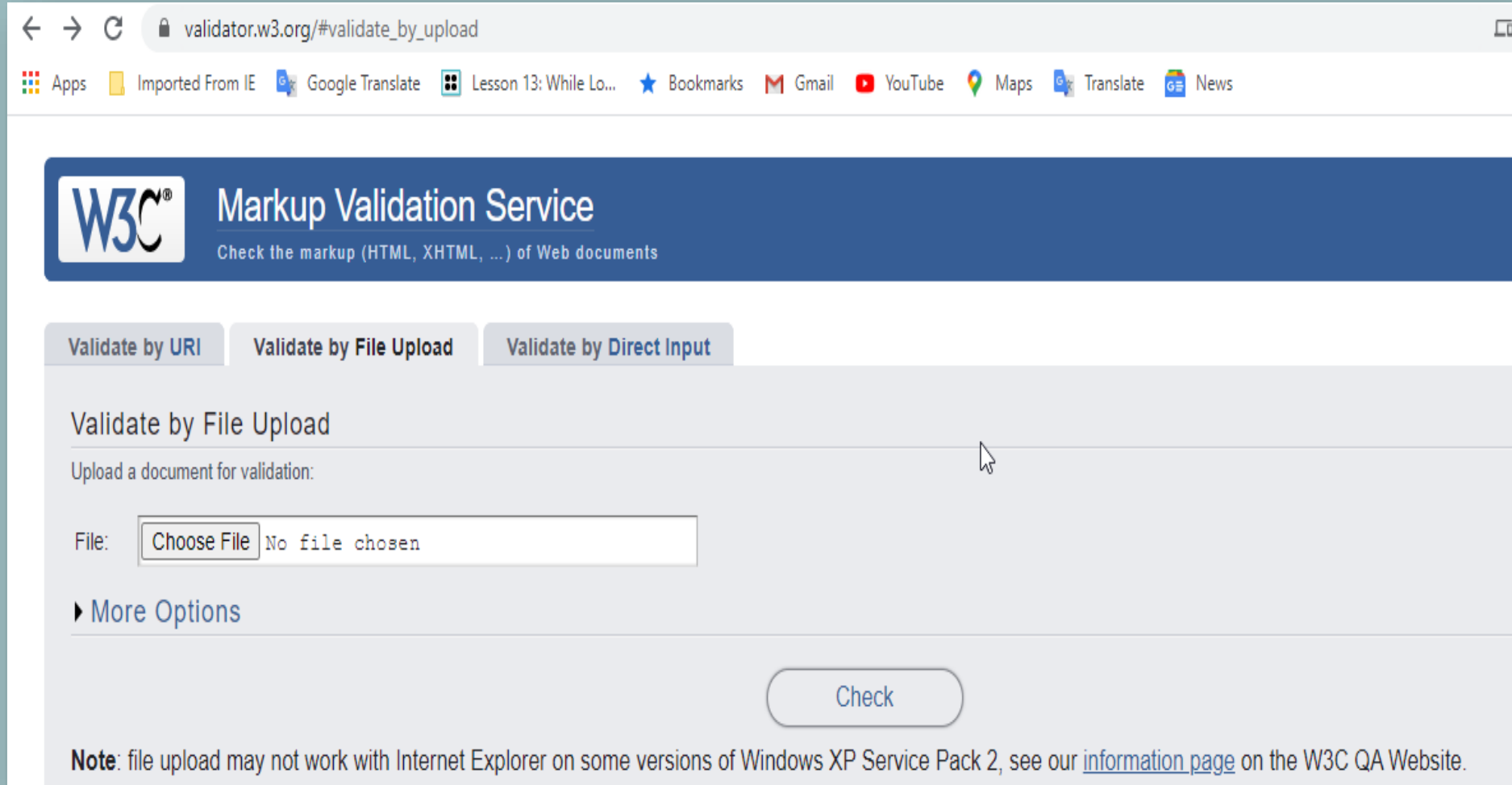
```
<!DOCTYPE html>
<html>
  <head>...</head>
  <body>
    <div id="tips">...</div>
     == $0
    <br>
    
    <br>
    <a href="#tips">go to tips section</a>
    <!-- Code injected by live-server -->
    <script type="text/javascript">...</script>
  </body>
</html>
```

The **Styles** panel on the right shows the default styles for the selected **img** element:

```
element.style {
}
img[Attributes Style] {
  width: 50%;
  height: 50%;
}
```

The breadcrumb at the bottom of the developer tools indicates the element's path: **html > body > img**.

https://validator.w3.org



The screenshot shows a web browser window with the URL `validator.w3.org/#validate_by_upload`. The browser's address bar and tabs are visible at the top. The main content area features the W3C logo and the title "Markup Validation Service" with the subtitle "Check the markup (HTML, XHTML, ...) of Web documents". Below this, there are three tabs: "Validate by URI", "Validate by File Upload" (which is selected), and "Validate by Direct Input". The "Validate by File Upload" section contains the text "Validate by File Upload" and "Upload a document for validation:". Below this is a file selection interface with a "File:" label, a "Choose File" button, and the text "No file chosen". A "More Options" link is also present. At the bottom of the form is a large "Check" button. A note at the very bottom states: "Note: file upload may not work with Internet Explorer on some versions of Windows XP Service Pack 2, see our [information page](#) on the W3C QA Website."

← → ↻ `validator.w3.org/#validate_by_upload`

Apps Imported From IE Google Translate Lesson 13: While Lo... Bookmarks Gmail YouTube Maps Translate News

W3C® Markup Validation Service
Check the markup (HTML, XHTML, ...) of Web documents

Validate by URI Validate by File Upload Validate by Direct Input

Validate by File Upload

Upload a document for validation:

File: No file chosen

► More Options

Note: file upload may not work with Internet Explorer on some versions of Windows XP Service Pack 2, see our [information page](#) on the W3C QA Website.

Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for uploaded file index.html

Checker Input

Show ☐ source ☐ outline ☐ image report

Check by No file chosen

Uploaded files with .xhtml or .xht extensions are parsed using the XML parser.

Use the Message Filtering button below to hide/show particular messages, and to see total counts of errors and warnings.

1. **Warning** Consider adding a `lang` attribute to the `html` start tag to declare the language of this document.

From line 1, column 16; to line 2, column 6

TYPE `html>``<html>``<head`

For further guidance, consult [Declaring the overall language of a page](#) and [Choosing language tags](#).

If the HTML checker has misidentified the language of this document, please [file an issue report](#) or [send e-mail to report the problem](#).

2. **Error** The character encoding was not declared. Proceeding using `windows-1252`.

3. **Error** Bad value `50%` for attribute `width` on element `img`: Expected a digit but saw `%` instead.

From line 15, column 3; to line 15, column 67

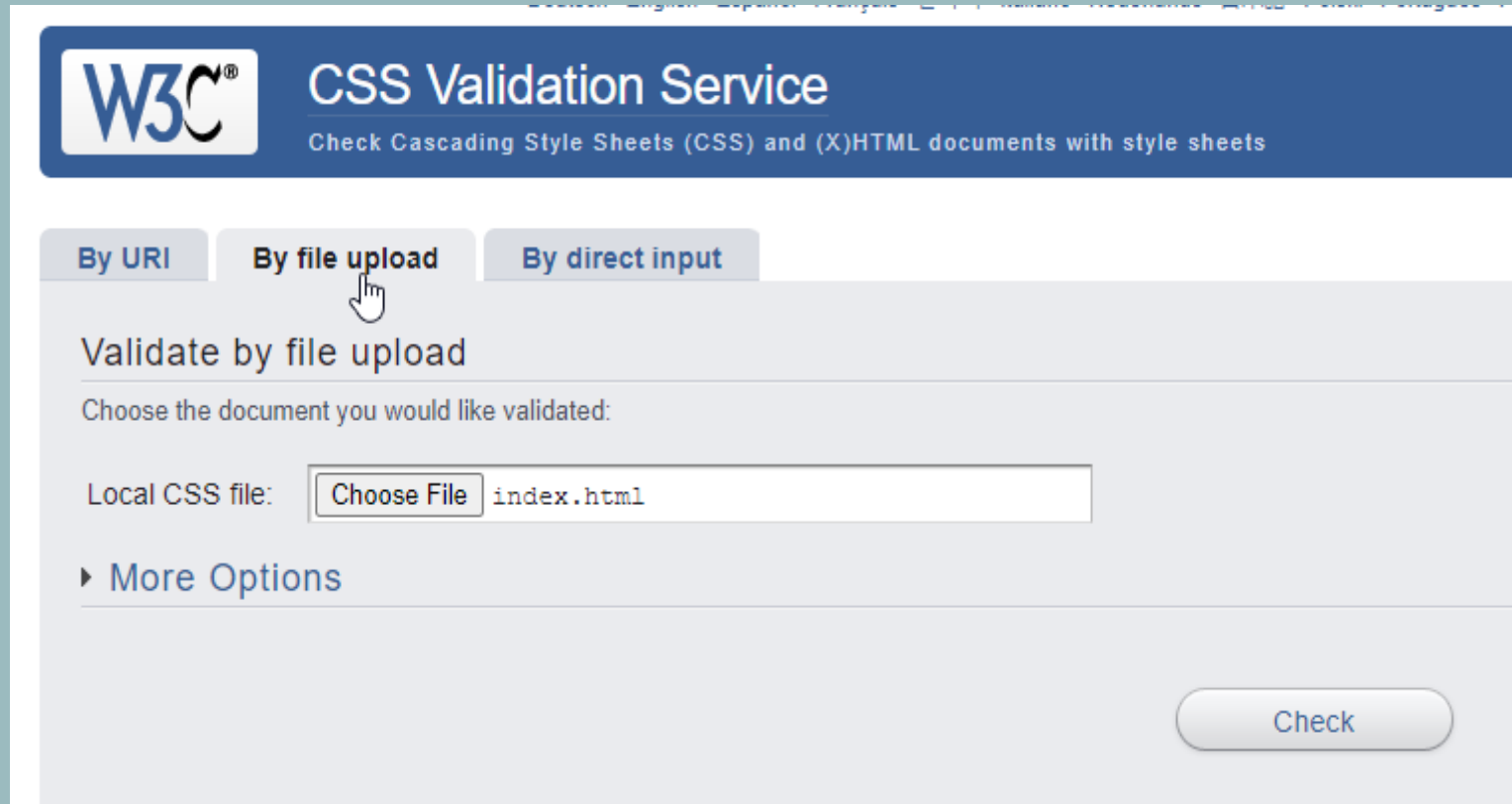
`</div>````<br`

4. **Error** Bad value `50%` for attribute `height` on element `img`: Expected a digit but saw `%` instead.

From line 15, column 3; to line 15, column 67

`</div>````<br`

<https://jigsaw.w3.org/css-validator>



The screenshot shows the W3C CSS Validation Service interface. At the top, there is a blue header with the W3C logo and the text "CSS Validation Service" and "Check Cascading Style Sheets (CSS) and (X)HTML documents with style sheets". Below the header, there are three tabs: "By URI", "By file upload", and "By direct input". The "By file upload" tab is selected, and a mouse cursor is pointing at it. Below the tabs, the text "Validate by file upload" is displayed. Underneath, it says "Choose the document you would like validated:". There is a label "Local CSS file:" followed by a text input field containing "index.html" and a "Choose File" button. Below this, there is a link "More Options" with a right-pointing triangle. At the bottom right, there is a "Check" button.

W3C[®] CSS Validation Service
Check Cascading Style Sheets (CSS) and (X)HTML documents with style sheets

By URI By file upload By direct input

Validate by file upload

Choose the document you would like validated:

Local CSS file: Choose File index.html

► More Options

Check



CSS Validation Service

Check Cascading Style Sheets (CSS) and (X)HTML documents with style sheets

By URI

By file upload

By direct input



Validate by direct input

Enter the CSS you would like validated:

```
table, th, td {  
  border: 1px solid black;  
  border-collapse: collapse;  
  padding: 15px;  
}  
table {  
  border-spacing: 5px;  
}  
th {  
  text-align: left;  
  background-color: #eee;  
}
```

► More Options

Check

W3C CSS Validator results for TextArea (CSS level 3 + SVG)

Congratulations! No Error Found.

This document validates as [CSS level 3 + SVG](#) !

To show your readers that you've taken the care to create an interoperable Web page, you may display this icon on any page that validates. Here is the XHTML you could use to add this icon to your Web page:



```
<p>
  <a href="http://jigsaw.w3.org/css-validator/check/referer">
    
  </a>
</p>
```



```
<p>
  <a href="http://jigsaw.w3.org/css-validator/check/referer">
    
  </a>
</p>
```

(close the img tag with > instead of /> if using HTML <= 4.01)